

# TCP/IP Stack Core Concepts

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| Cue /<br>Keywords | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| TCP vs IP         | - <b>TCP (Transmission Control Protocol)</b> : Manages data transmission; ensures reliable, ordered, error-checked delivery between applications.<br>- <b>IP (Internet Protocol)</b> : Core internet protocol; handles logical transmission, addressing, and routing via IP addresses.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| TCP/IP<br>Layers  | 1. <b>Application</b> : Network services directly to end-user apps (web, email). Generates data.<br>2. <b>Transport (TCP)</b> : End-to-end communication, data integrity, error recovery, flow control.<br>3. <b>Internet/Network (IP)</b> : Logical transmission, routing packets via IP addresses.<br>4. <b>Network Access / Data Link</b> : Error prevention and physical framing.                                                                                                                                                                                                                                                                                                                                                                                                |
| OSI<br>Mapping    | - Application (TCP/IP) -> Application, Presentation, Session (OSI)<br>- Transport (TCP/IP) -> Transport (OSI)<br>- Internet (TCP/IP) -> Network (OSI)<br>- Network Access (TCP/IP) -> Data Link, Physical (OSI)<br>- <b>Layers</b> : TCP/IP (4 layers) combines Session & Presentation into Application; OSI (7 layers) separates them.<br>- <b>Differences</b> : <b>Approach</b> : TCP/IP is horizontal/connectionless; OSI is vertical.<br>- <b>Delivery</b> : TCP/IP does not guarantee packet delivery by default; OSI features guaranteed delivery.<br>- <b>Network Layer</b> : TCP/IP is connectionless (IP) and relies on Transport (TCP) for connections; OSI Network provides both.<br>- <b>Replaceability</b> : OSI protocols are easier to replace than TCP/IP protocols. |

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## Summary

The TCP/IP stack is a foundational 4-layer model (Application, Transport, Internet, Network Access) that governs internet communication. It takes a more horizontal, connectionless approach compared to the traditional 7-layer OSI model, heavily relying on IP for routing and TCP for reliable end-to-end delivery.